# Investigating with commands

**NUnix · L3 MIAGE · CM2 · 90 minutes**

ISTIC · University of Rennes 2026–2027

Université de Rennes
Louis Ledoux



## Investigating with commands

Université de Rennes
Louis Ledoux

A computer is slow. Would adding RAM help?
Check which resource is under pressure first.

We will use commands to inspect the system, find useful information, and turn it into a summary we can save.



## CM2 outline

Université de Rennes
Louis Ledoux

1. Processes and resources
2. Finding files and text
3. Inputs, outputs and redirections
4. Combining commands with pipes

Observe the system → find information → redirect streams → combine commands.



Université de Rennes
Louis Ledoux

## 1. Processes and resources

Which programs are running, and where are the resources going?



## CPU, memory and disk space

Université de Rennes
Louis Ledoux

| Resource | Question | First observation |
|---|---|---|
| **CPU** | Is computation keeping a processor busy? | `top`, `ps` |
| **RAM** | Is memory pressure affecting the system? | `free -h` |
| **Disk** | Can the filesystem store more data? | `df -h .` |

Adding RAM does not create free space on a full disk.



## Running in the foreground

Université de Rennes
Louis Ledoux

A **process** is a running instance of a program.

```
sleep 10
```

The shell waits for this foreground command to finish before showing a new prompt.

**Ctrl+C** requests an interrupt of the foreground job.

## Inspecting processes

Université de Rennes
Louis Ledoux

```
ps
ps -ef
```

| Field | Meaning |
|---|---|
| PID | Process identifier |
| TTY | Associated terminal, if any |
| TIME | Accumulated CPU time |
| CMD | Command |

7/50

## Checking memory usage

Université de Rennes
Louis Ledoux

```
free -h
ps -eo pid,comm,%cpu,%mem --sort=-%mem
```

- Read **available** memory; cache use is normal.
- Identify the application's process and follow it over time.
- Check logs before attributing every delay to memory.

8/50

## Checking disk usage

Université de Rennes
Louis Ledoux

| Tool | Scope | Question |
|---|---|---|
| df | A filesystem | How much space is available? |
| du | A directory tree | Which directories use space? |
| find | Individual entries | Which files fit my criteria? |

Start broad, then inspect a **specific file**.

9/50

## Find which directories use space

Université de Rennes
Louis Ledoux

```
df -h "$HOME"
du -h --max-depth=1 "$HOME"
```

df shows usage of the filesystem containing your home.
du shows how much space your directories use.

10/50

## From resources to files

Université de Rennes
Louis Ledoux

1. Processes and resources
2. **Finding files and text**
3. Inputs, outputs and redirections
4. Combining commands with pipes

We can inspect CPU, memory and disk usage. Now we need to locate files and find the relevant lines in their contents.

11/50

Université de Rennes
Louis Ledoux

# 2. Finding files and text

Where is the file, and which lines contain the information we need?

12/50

## Finding files

Université de Rennes
Louis Ledoux

```
find . -type f -name '*.txt'
```

- `.` is the **starting directory**.
- `find` also visits its subdirectories.
- `-type f` selects regular files.
- `-name '*.txt'` tests each entry's name.

The quotes let **find** receive the pattern unchanged.

13/50

## Search criteria

Université de Rennes
Louis Ledoux

| Criterion | Matches |
|---|---|
| `-iname '*.txt'` | Name, ignoring case |
| `-user alice` | Entries owned by Alice |
| `-type d` / `-type l` | Directories / symbolic links |
| `-size +100M` | Size greater than 100 MiB |
| `-mtime -2` | Modified less than 48 hours ago |

```
find . -type f -size +100M
```

14/50

## Checkpoint: find

Université de Rennes
Louis Ledoux

1. Find regular `.c` files below the current directory.
2. Why quote `'*.c'`?

1. `find . -type f -name '*.c'`
2. So the shell passes the pattern to `find` without expanding it first.

15/50

## A small log file

Université de Rennes
Louis Ledoux

We have located a log file. Next, we will search its contents with `grep`. Save these four lines as `app.log`:

```
10:00 INFO STARTED
10:01 ERROR DISK_FULL
10:02 ERROR TIMEOUT
10:03 ERROR DISK_FULL
```

| Field | Meaning |
|---|---|
| 1 | Time |
| 2 | Severity |
| 3 | Event code |

16/50

## Finding text with grep

Université de Rennes
Louis Ledoux

```
grep ' ERROR ' app.log
grep -r ' ERROR ' .
grep -n 'DISK_FULL' app.log
```

- Print **matching lines** from a file.
- `-r` searches directories recursively.
- `-n` includes line numbers.

Exit status: **0** = match, **1** = no match, **2** = error in GNU `grep`.

17/50

## Where the name grep comes from

Université de Rennes
Louis Ledoux

The editor ed had the command pattern:

```
g/re/p
```

- **g**: apply globally to matching lines.
- **re**: a regular expression.
- **p**: print those lines.

`grep` makes that text search available as a separate command.

18/50

## Patterns in grep

Université de Rennes
Louis Ledoux

| Regular expression | Meaning |
|---|---|
| `.` | Any one character |
| `m*` | Zero or more repetitions of `m` |
| `[1-3]` | A digit from 1 to 3 |

```
grep 'm.*t' names.txt
grep 'toto[1-3]' names.txt
```

In a regular expression, * repeats the **preceding item**.

19/50

## Who interprets this pattern?

Université de Rennes
Louis Ledoux

```
printf '%s\n' *.log
grep 'DISK.*' app.log
find . -name '*.log'
```

1. The **shell** expands a filename pattern.
2. **grep** interprets a regular expression.
3. **find** interprets a quoted filename pattern.

20/50

## From matching lines to saved results

Université de Rennes
Louis Ledoux

1. Processes and resources
2. Finding files and text
3. **Inputs, outputs and redirections**
4. Combining commands with pipes

We can find files and select useful lines. How can a command read from a file and save its output or errors?

21/50

Université de Rennes
Louis Ledoux

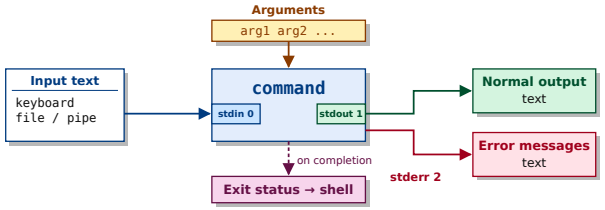
# 3. Inputs, outputs and redirections

Where does a command read, where does it write, and did it succeed?

22/50

## Anatomy of a command

Université de Rennes
Louis Ledoux




**Arguments** configure the command. **Streams** carry text. The **exit status** reports success or failure when the command finishes.

23/50

## Exit status

Université de Rennes
Louis Ledoux

```
true
echo "$?"
false
echo "$?"
```

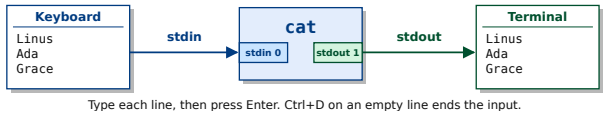
**0** means success / true. **Nonzero** means failure / false.
$? is the status of the **most recent command**.

24/50

## Keyboard input and screen output

Université de Rennes
Louis Ledoux

```
cat
```




Type each line, then press Enter. Ctrl+D on an empty line ends the input.

With no file argument, `cat` copies **stdin** to **stdout**.

25/50

## Reading input from a file

Université de Rennes
Louis Ledoux

```
sort < names.txt
```



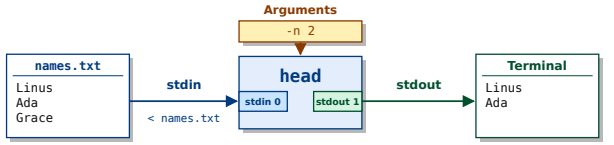

`< names.txt` makes the shell connect the file to **stdin**. The sorted lines still go to the terminal through **stdout**.

26/50

## Arguments and input

Université de Rennes
Louis Ledoux

```
head -n 2 < names.txt
```




`-n 2` tells head **how many lines** to keep. `< names.txt` supplies **the text** to read.

27/50

## Saving output

Université de Rennes
Louis Ledoux

```
sort < names.txt > sorted.txt
```




`> sorted.txt` connects **stdout** to a file. The shell creates the file or replaces its contents.

28/50

## Appending to a file

Université de Rennes
Louis Ledoux

```
echo 'Zoé' >> sorted.txt
```

| Operator | Effect on the destination file |
|---|---|
| `>` | Create it, or **replace** its contents |
| `>>` | Create it, or **append** to its contents |

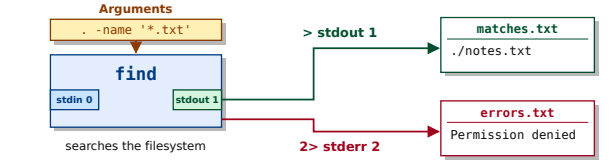
Zoé is added after the existing lines in `sorted.txt`.

29/50

## Redirecting errors

Université de Rennes
Louis Ledoux

```
find . -name '*.txt' > matches.txt 2> errors.txt
```




`>` sends normal output to `matches.txt`. `2>` sends error messages to `errors.txt`.

30/50

## Redirections are applied from left to right

Université de Rennes
Louis Ledoux

```
ls notes.txt missing > all.txt 2>&1
ls notes.txt missing 2>&1 > out.txt
```

| Command | stdout ends up… | stderr ends up… |
|---|---|---|
| First | In `all.txt` | In `all.txt` |
| Second | In `out.txt` | At the terminal |

2>&1 copies stdout's destination **at that moment**.
It does not make stderr follow later changes to stdout.

31/50

## Why did the file become empty?

Université de Rennes
Louis Ledoux

```
sort names.txt > names.txt
```

The shell opens and truncates `names.txt` **before** starting `sort`. The input is already gone.

Write to a **different file**, then inspect the result.

32/50

## From one command to a pipeline

Université de Rennes
Louis Ledoux

1. Processes and resources
2. Finding files and text
3. Inputs, outputs and redirections
4. **Combining commands with pipes**

We can choose the input and output of one command. Now we will connect commands to count and rank the errors in our log.

33/50

Université de Rennes
Louis Ledoux

# 4. Combining commands with pipes

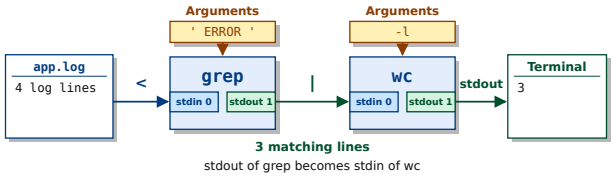
How can one command use another command's output?

34/50

## A pipe between two commands

Université de Rennes
Louis Ledoux

```
grep ' ERROR ' < app.log | wc -l
```




| connects the first command's **stdout** to the next command's **stdin**.
`wc -l` counts the matching lines: **3** for our sample log.

35/50

## Connecting programs like garden hoses

Université de Rennes
Louis Ledoux

Doug McIlroy described connecting programs so that one program's output could become another's input.

```
grep ' ERROR ' app.log | wc -l
```

Each tool has a small job.
The shared text stream lets us combine them.

36/50

## Pipeline step 1: extract error codes

Université de Rennes
Louis Ledoux

```
grep ' ERROR ' app.log |
    cut -d ' ' -f 3
```

```
DISK_FULL
TIMEOUT
DISK_FULL
```

37/50

## Pipeline step 2: count each code

Université de Rennes
Louis Ledoux

```
grep ' ERROR ' app.log |
    cut -d ' ' -f 3 |
    sort | uniq -c | sort -nr
```

```
      2 DISK_FULL
      1 TIMEOUT
```

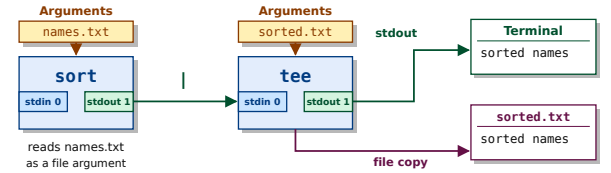
`uniq -c` counts **adjacent** equal lines. Sort first.

38/50

## Save and display output with tee

Université de Rennes
Louis Ledoux

```
sort names.txt | tee sorted.txt
```



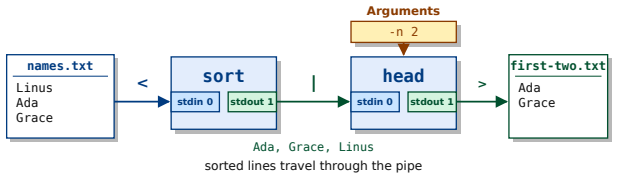

Like a **T-shaped pipe fitting**, tee copies the text to a file and to **stdout**.

39/50

## Chaining commands with arguments

Université de Rennes
Louis Ledoux

```
sort < names.txt | head -n 2 > first-two.txt
```




`<` chooses the input file. `|` joins the commands. `>` chooses the output file. `-n 2` remains an argument to head.

40/50

## Checkpoint: redirections

Université de Rennes
Louis Ledoux

1. Which operator appends to a file?
2. What does a `|` b connect?
3. Does a `> out.txt` also redirect standard error?

1. `>>`
2. a's stdout to b's stdin.
3. No. Use `2>` to redirect stderr.

41/50

## The exit status of a pipeline

Université de Rennes
Louis Ledoux

```
false | sort
echo "$?"
set -o pipefail
false | sort
echo "$?"
```

Output: **0**, then **1**.
Bash normally reports the last pipeline command's status.

42/50

## CM2 summary: commands and pipelines

Université de Rennes
Louis Ledoux

1. **Processes and resources:** inspect CPU, memory and disk usage.
2. **Finding files and text:** locate files with `find`, select lines with `grep`.
3. **Inputs, outputs and redirections:** read files, save results, separate errors.
4. **Combining commands with pipes:** extract, count and rank log entries.

43/50

Université de Rennes
Louis Ledoux

## Optional reference

More search expressions and interactive process control

44/50

## Combining criteria

Université de Rennes
Louis Ledoux

```
find /usr -name '*.c'
find . -user alice -name '*.c'
find . -not \( -name '*.obj' -o -name '*.map' \)
```

- Adjacent tests use **AND**. `-a` makes it explicit.
- `-o` means **OR**. `-not` negates a test.
- Escaped parentheses group tests for `find`.

45/50

## Running a command on matches

Université de Rennes
Louis Ledoux

```
find . -type f -name '*.txt' -exec wc -l {} \;
```

- `-exec` starts the command to run.
- `{}` stands for the current matching path.
- `\;` ends the command without ending the shell's command line.

46/50

## Running in the background

Université de Rennes
Louis Ledoux

```
sleep 60 &
jobs
fg %1
```

- `&` lets the shell return to the prompt immediately.
- `jobs` lists jobs managed by this shell.
- `fg %1` brings job **1** to the foreground.

47/50

## Suspending and resuming a job

Université de Rennes
Louis Ledoux

| Action | Effect |
|---|---|
| **Ctrl+Z** | Suspend the foreground job |
| `bg %1` | Resume job 1 in the background |
| `fg %1` | Resume job 1 in the foreground |

A **job number** belongs to a shell.
A **PID** identifies a process in the system.

48/50

## Sequential and asynchronous execution

Université de Rennes
Louis Ledoux

```
who | sort; date
sleep 10 &
date
```

- `;` runs the next command after the previous one finishes.
- `&` starts work asynchronously.
- The final `date` can run while `sleep` is still active.

49/50

## Signals and end of input

Université de Rennes
Louis Ledoux

| Action | Meaning |
|---|---|
| `kill PID` | Send `SIGTERM` to a process |
| **Ctrl+C** | Send `SIGINT` to the foreground process group |
| `Ctrl+\` | Send `SIGQUIT` to the foreground process group |
| **Ctrl+Z** | Send `SIGTSTP` to suspend the foreground group |
| **Ctrl+D** | Terminal end-of-input action |

50/50